

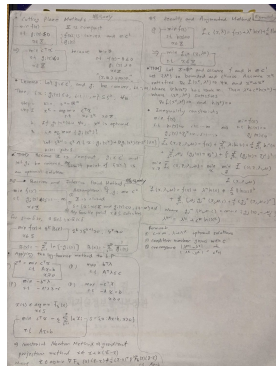
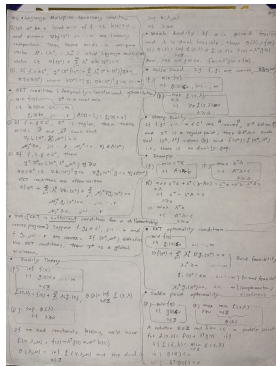
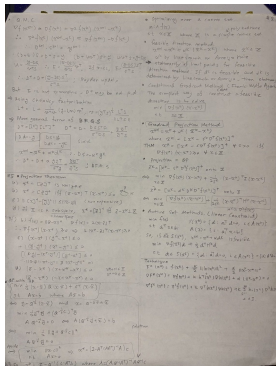
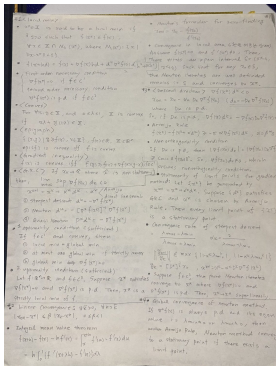
Nonlinear Programming — Summary Notes

Cheat sheet for David Morton's Nonlinear Programming course, UT Austin

Originally summarized by Hongseok Kim as a first-year PhD student in Spring 2006

Reconstructed as slides, 2026

The original cheat sheet (2006)



Four handwritten pages, sections #1-#9. Everything that follows is a typeset reconstruction of these.

Outline

- 1 Unconstrained optimization
- 2 Quasi-Newton methods
- 3 Optimization over a convex set
- 4 Lagrange multipliers, KKT, and duality
- 5 Algorithms for constrained problems

Local minima and optimality conditions

Local minimum

$x^* \in X$ is a *local min* if $\exists \varepsilon > 0$ s.t. $f(x^*) \leq f(x)$ for all $x \in X \cap N_\varepsilon(x^*)$, where $N_\varepsilon(x^*) = \{x : \|x - x^*\| < \varepsilon\}$.

Taylor expansion

$$f(x + hd) = f(x) + h \nabla f(x)^T d + \frac{h^2}{2} d^T \nabla^2 f(x) d + o(h^2)$$

Necessary conditions

- First order: $\nabla f(x^*) = 0$ ($f \in C^1$)
- Second order: $\nabla^2 f(x^*)$ is p.s.d. ($f \in C^2$)

Sufficient condition (second order)

$f : \mathbb{R}^n \rightarrow \mathbb{R}$, $f \in C^2$. If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is p.d., then x^* is a strict local min.

Convexity

- **Convex set:** X is convex if $\lambda x + (1 - \lambda)y \in X$ for all $x, y \in X$, $0 \leq \lambda \leq 1$.
- **Epigraph:** $\text{epi}(f) = \{(x, y) : y \geq f(x), x \in X\}$, $f : X \rightarrow \mathbb{R}$, $X \subseteq \mathbb{R}^n$.
 $\text{epi}(f)$ is convex $\iff f$ is convex.
- **Gradient inequality** ($f \in C^1$):

$$f \text{ convex} \iff f(y) \geq f(x) + \nabla f(x)^T(y - x) \quad \forall x, y.$$

Sufficient conditions under convexity ($f \in C^1$, convex)

- 1 local min = global min
- 2 at most one global min if f is strictly convex
- 3 x^* global min $\iff \nabla f(x^*) = 0$

Descent methods

$$x^{k+1} = x^k + \alpha^k d^k, \quad \alpha^k : \text{Armijo rule or direct line search}$$

Search directions

- ① Steepest descent: $d^k = -\nabla f(x^k)$
- ② Newton: $d^k = -[\nabla^2 f(x^k)]^{-1} \nabla f(x^k)$
- ③ Quasi-Newton: $D^k d^k = -\nabla f(x^k)$

Descent direction: $\nabla f(x^k)^\top d^k < 0$.

With $d^k = -D_k \nabla f(x^k)$, D_k p.d.:

$$\nabla f(x^k)^\top d^k = -\nabla f(x^k)^\top D_k \nabla f(x^k) < 0.$$

Armijo rule

Choose $\alpha = \beta^m$ with the smallest $m \geq 0$ such that

$$f(x^k) - f(x^k + \alpha d^k) \geq -\sigma \alpha \nabla f(x^k)^\top d^k.$$

Gradient-related directions and stationarity of limit points

Gradient-related condition (G.R.C.)

If $x^k \rightarrow \hat{x}$ with $\hat{x}$ nonstationary, then $\limsup_{k \rightarrow \infty} \nabla f(x^k)^\top d^k < 0$.

Non-orthogonality

If D_k is p.d. with eigenvalues bounded below by $c_{\min} > 0$,

$$|\nabla f(x^k)^\top d^k| = |\nabla f(x^k)^\top D_k \nabla f(x^k)| \geq c_{\min} \|\nabla f(x^k)\|^2,$$

so $\nabla f(x^k)^\top d^k \neq 0$ whenever $\nabla f(x^k) \neq 0$.

Theorem (stationarity of limit points)

Let $\{x^k\}$ be generated by $x^{k+1} = x^k + \alpha^k d^k$. If $\{d^k\}$ satisfies G.R.C. and α^k is chosen by the Armijo rule, then every limit point of $\{x^k\}$ is a stationary point.

Convergence rates

Linear convergence

$\exists \bar{k}, 0 \leq \beta < 1: \|x_{k+1} - x^*\| \leq \beta \|x_k - x^*\|$ for all $k > \bar{k}$.

Steepest descent on a quadratic (scaled)

With $y_k = [D^k]^{1/2} x_k$ and $x^{k+1} = x^k - \alpha^k D^k \nabla f(x^k)$,

$$\frac{\|y_{k+1}\|}{\|y_k\|} \leq \max\{|1 - \alpha^k \lambda_{\min}|, |1 - \alpha^k \lambda_{\max}|\}.$$

Best constant step $\alpha = \frac{2}{\lambda_{\max} + \lambda_{\min}}$ gives rate $\frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}}$.

Newton's method: local and global convergence

Newton's method

- Local: $f \in C^2$, pure Newton iterates converge to x^* with $\nabla f(x^*) = 0$, $\nabla^2 f(x^*)$ p.d. $\Rightarrow x^k \rightarrow x^*$ superlinearly.
- Global: if $\nabla^2 f(x)$ is p.d. everywhere with eigenvalues bounded ($\lambda_{\min} \geq c > 0$, $\lambda_{\max}$ bounded), then under the Armijo rule Newton's method converges to a stationary point whenever a limit point exists.

Scalar Newton for zero-finding

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

Local convergence ($f \in C^1$)

Assume $f(x^*) = 0$ and $f'(x^*) \neq 0$. Then there is an open interval $S = (x^* - \varepsilon, x^* + \varepsilon)$ such that for any $x_0 \in S$ the Newton iterates are well defined, remain in S , and converge to x^* .

Integral mean value theorem (tool for the proof)

$$f(x+h) - f(x) - hf'(x) = \int_x^{x+h} [f'(u) - f'(x)] du = h \int_0^1 [f'(x + \lambda h) - f'(x)] d\lambda.$$

Secant condition and Broyden's rank-one update

Quasi-Newton (secant) condition

$\nabla f(x^{k+1}) \approx \nabla f(x^k) + \nabla^2 f(x^k)(x^{k+1} - x^k)$, hence require

$$D^{k+1}s^{k+1} = y^{k+1}, \quad s^{k+1} = x^{k+1} - x^k, \quad y^{k+1} = \nabla f(x^{k+1}) - \nabla f(x^k).$$

Rank-one correction $D^+ = D + E$, $E = uv^T$

$(D + uv^T)s = y \Rightarrow u(v^T s) = y - Ds \Rightarrow u = \frac{y - Ds}{v^T s}$, so $E = \frac{(y - Ds)v^T}{v^T s}$. Taking $v = s$:

$$D^+ = D + \frac{(y - Ds)s^T}{s^T s} \quad (\text{Broyden update})$$

Drawback: E is not symmetric, so D^+ need not be p.d..

Cholesky-based update and BFGS

Update the Cholesky factor $D = LL^T$

$$L^+ = L + \frac{1}{v^T v} (y - Lv) v^T, \quad v = (y^T s)^{1/2} \frac{L^T s}{\|L^T s\|}, \quad D^+ = L^+ (L^+)^T.$$

Resulting BFGS update

$$D^+ = D - \frac{D s s^T D}{s^T D s} + \frac{y y^T}{y^T s}$$

Using $Dd = -g$, $s = \alpha d$ (so $Ds = -\alpha g$):

$$D^+ = D + \alpha \frac{g g^T}{s^T g} + \frac{y y^T}{y^T s}, \quad g = \nabla f(x^k).$$

Symmetric, and p.d. is preserved when $y^T s > 0$ (guaranteed by a Wolfe-type line search).

Projection theorem

Let $X \subseteq \mathbb{R}^n$ be closed convex and $[z]^+ = \arg \min_{x \in X} \|z - x\|$.

- (a) $x^* = [z]^+$ exists and is unique.
- (b) $x^* = [z]^+ \iff (z - x^*)^T(x - x^*) \leq 0 \quad \forall x \in X$.
- (c) $\|[x]^+ - [y]^+\| \leq \|x - y\|$ (nonexpansive).
- (d) If X is a subspace: $x^* = [z]^+ \iff z - x^* \perp X$.

Proof sketches

- (b) $f(x) = \|z - x\|^2$, $\nabla f(x) = 2(x - z)$; optimality over X : $\nabla f(x^*)^T(x - x^*) \geq 0$, i.e. $(x^* - z)^T(x - x^*) \geq 0$.
- (c) Add $(x - x^+)^T(y^+ - x^+) \leq 0$ and $(y - y^+)^T(x^+ - y^+) \leq 0$:
 $[(x - y) - (x^+ - y^+)]^T(y^+ - x^+) \leq 0 \Rightarrow \|x^+ - y^+\|^2 \leq \|x - y\| \|x^+ - y^+\|$.
- (d) Apply (b) with $x^* \pm x \in X$: $(z - x^*)^T x \leq 0$ and $-(z - x^*)^T x \leq 0$.

Equality-constrained QP: closed form

$$\min \frac{1}{2}(x - \bar{x})^T Q(x - \bar{x}) + c^T(x - \bar{x}) \quad \text{s.t. } Ax = b, \quad A\bar{x} = b.$$

Substitute $y = Q^{1/2}(x - \bar{x})$, $x = Q^{-1/2}y + \bar{x}$:

$$\min \frac{1}{2}y^T y + (Q^{-1/2}c)^T y \quad \text{s.t. } AQ^{-1/2}y = 0 \iff \min \frac{1}{2}\|y + Q^{-1/2}c\|^2 \quad \text{s.t. } AQ^{-1/2}y = 0.$$

Aside: projection onto the null space of A

$$\min \|x - c\|^2 \quad \text{s.t. } Ax = 0 \Rightarrow x^* = (I - A^T(AA^T)^{-1}A)c.$$

Solution

$$x^* = \bar{x} - Q^{-1}(c - A^T\lambda), \quad \lambda = (AQ^{-1}A^T)^{-1}AQ^{-1}c.$$

Feasible direction methods and conditional gradient

$\min f(x)$ s.t. $x \in X$, X a simple convex set (e.g. polyhedron).

Feasible direction method

$$x^{k+1} = x^k + \alpha^k(\bar{x}^k - x^k), \quad \bar{x}^k \in X, \quad \alpha^k \in (0, 1] \text{ by line search or Armijo.}$$

If $d^k = \bar{x}^k - x^k$ is feasible and gradient-related, and α^k is chosen by line search or Armijo, then every limit point is stationary.

Conditional gradient (Frank–Wolfe)

The simplest way to construct a feasible direction:

$$\bar{x}^k \in \arg \min_{x \in X} \nabla f(x^k)^T (x - x^k).$$

Gradient projection method

$$x^{k+1} = x^k + \alpha^k(\bar{x}^k - x^k), \quad \bar{x}^k = [x^k - s^k \nabla f(x^k)]^+ \text{ (projection onto } X\text{)}.$$

Theorem (fixed point characterization)

$$x^* = [x^* - s \nabla f(x^*)]^+ \text{ for all } s > 0 \iff \nabla f(x^*)^\top (x - x^*) \geq 0 \text{ for all } x \in X.$$

Projection as a QP

$$\bar{x}^k = [x^k - s^k \nabla f(x^k)]^+ \iff \bar{x}^k = \arg \min_{x \in X} \nabla f(x^k)^\top (x - x^k) + \frac{1}{2s^k} (x - x^k)^\top (x - x^k).$$

Scaled version ($H^k \approx \nabla^2 f(x^k)$):

$$\bar{x}^k = \arg \min_{x \in X} \nabla f(x^k)^\top (x - x^k) + \frac{1}{2s^k} (x - x^k)^\top H^k (x - x^k).$$

Active set methods (linear constraints)

$$\min f(x) \quad \text{s.t.} \quad a_i^T x \leq b_i, \quad i = 1, \dots, r.$$

- Active set: $A(x) = \{i : a_i^T x = b_i\}$.
- Feasible direction subspace at x^k : $S(x^k) = \{d : a_i^T d = 0, \quad i \in A(x^k)\} = \{d : A^k d = 0\}$.
- If $d^k \in S(x^k)$, then $x^{k+1} = x^k + \alpha d^k$ stays feasible for small α .

Direction subproblem

$$\min_d \quad \nabla f(x^k)^T d + \frac{1}{2} d^T H^k d \quad \text{s.t.} \quad A^k d = 0.$$

(an equality-constrained QP; see the closed form above)

Lagrange multiplier necessary conditions (equality constraints)

$$\min f(x) \text{ s.t. } h(x) = 0, h : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

First order

Let x^* be a local min and $\nabla h_1(x^*), \dots, \nabla h_m(x^*)$ linearly independent (regular point). Then there exists a unique $\lambda^* = (\lambda_1^*, \dots, \lambda_m^*)$ with

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla h_i(x^*) = 0.$$

Second order ($f, h \in C^2$)

$$y^T \left(\nabla^2 f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla^2 h_i(x^*) \right) y \geq 0 \quad \forall y \in V(x^*) = \{y : \nabla h_i(x^*)^T y = 0, i = 1, \dots, m\}.$$

KKT necessary conditions (inequality-constrained)

$$\min f(x) \quad \text{s.t.} \quad h_i(x) = 0 \quad (i = 1, \dots, m), \quad g_j(x) \leq 0 \quad (j = 1, \dots, r), \quad A(x) = \{j : g_j(x) = 0\}.$$

$L(x, \lambda, \mu) = f(x) + \lambda^T h(x) + \mu^T g(x)$. Let x^* be a local min and a regular point.

① $(f, h, g \in C^1)$ There exist λ^*, μ^* such that

$$\nabla_x L(x^*, \lambda^*, \mu^*) = 0, \quad \mu_j^* \geq 0, \quad \mu_j^* = 0 \quad \forall j \notin A(x^*).$$

② $(f, h, g \in C^2)$ $y^T \nabla_{xx}^2 L(x^*, \lambda^*, \mu^*) y \geq 0$ for all y with $\nabla h_i(x^*)^T y = 0$ and $\nabla g_j(x^*)^T y = 0$, $j \in A(x^*)$.

Usual form

$$\nabla f(x^*) + \sum_i \lambda_i^* \nabla h_i(x^*) + \sum_j \mu_j^* \nabla g_j(x^*) = 0, \quad \mu_j^* g_j(x^*) = 0, \quad \mu_j^* \geq 0.$$

KKT as a sufficient condition; the three ingredients

Theorem (differentiable convex program)

Suppose $f, g_j \in C^1$ and f, g_j convex, $j = 1, \dots, r$. If (x^*, μ^*) satisfies the KKT conditions, then x^* is a global minimum.

KKT for $\min f(x)$ s.t. $g_i(x) \leq 0, i = 1, \dots, m$

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x^*) = 0, \quad \lambda_i^* \geq 0 \quad \text{dual feasibility}$$

$$g_i(x^*) \leq 0 \quad \text{primal feasibility}$$

$$\lambda_i^* g_i(x^*) = 0 \quad \text{complementary slackness}$$

Lagrangian duality and weak duality

$$\textbf{(P)} \quad \inf f(x) \text{ s.t. } g_i(x) \leq 0, x \in X$$

$$L(x, \lambda) = f(x) + \sum_i \lambda_i g_i(x), \quad \theta(\lambda) = \inf_{x \in X} L(x, \lambda)$$

$$\textbf{(D)} \quad \sup \theta(\lambda) \text{ s.t. } \lambda \geq 0$$

With equality constraints $h(x) = 0$ as well:

$$L(x, \lambda, \mu) = f + \lambda^T g + \mu^T h, \quad \theta(\lambda, \mu) = \inf_{x \in X} L, \\ \text{dual: } \sup \theta(\lambda, \mu) \text{ s.t. } \lambda \geq 0 \text{ } (\mu \text{ free}).$$

Weak duality

If x is primal feasible and λ is dual feasible, then $\theta(\lambda) \leq f(x)$.

$$\theta(\lambda) = \inf_{x' \in X} L(x', \lambda) \leq L(x, \lambda) = f(x) + \lambda^T g(x) \leq f(x) \quad (\lambda \geq 0, g(x) \leq 0).$$

Wolfe dual and strong duality

Wolfe dual (for f, g_i convex and differentiable)

$$(P) \min f(x) \text{ s.t. } g_i(x) \leq 0 \quad (D) \max_{x, \lambda} L(x, \lambda) \text{ s.t. } \nabla_x L(x, \lambda) = 0, \lambda \geq 0.$$

Strong duality

If $f, g_i \in C^1$ are convex, x^* solves (P) and x^* is a regular point, then there exists $\lambda^* \geq 0$ such that (x^*, λ^*) solves (D) and

$$f(x^*) = L(x^*, \lambda^*),$$

i.e. there is no duality gap.

Example: LP duality via the Wolfe dual

$$(P) \min c^T x \quad \text{s.t.} \quad Ax \geq b$$

$$(D) \max b^T \lambda \quad \text{s.t.} \quad A^T \lambda = c, \lambda \geq 0.$$

Derivation

$L(x, \lambda) = c^T x + \lambda^T (b - Ax)$. Wolfe dual:

$$\max_{x, \lambda} c^T x + \lambda^T b - \lambda^T Ax \quad \text{s.t.} \quad c - A^T \lambda = 0, \lambda \geq 0.$$

Using the constraint, $c^T x - \lambda^T Ax = (c - A^T \lambda)^T x = 0$, so the objective reduces to $b^T \lambda$:

$$\max b^T \lambda \quad \text{s.t.} \quad A^T \lambda = c, \lambda \geq 0.$$

Saddle point optimality

$$(P) \min f(x) \text{ s.t. } g(x) \leq 0, x \in X \quad (D) \max_{\lambda \geq 0} \underbrace{\min_{x \in X} L(x, \lambda)}_{\theta(\lambda)}$$

Saddle point

$\bar{x} \in X$ and $\bar{\lambda} \geq 0$ form a saddle point of $L(x, \lambda) = f(x) + \lambda^T g(x)$ iff

- (i) $L(\bar{x}, \bar{\lambda}) = \min_{x \in X} L(x, \bar{\lambda})$
- (ii) $g(\bar{x}) \leq 0$
- (iii) $\bar{\lambda}^T g(\bar{x}) = 0$

In that case $\bar{x}$ solves (P), $\bar{\lambda}$ solves (D), and there is no duality gap.

Cutting plane method

$\min f(x)$ s.t. $g_i(x) \leq 0$, $x \in X$ compact; f, g_i convex and C^1 . Reduce to a linear objective in (x, θ) space: $\min \theta$ s.t. $f(x) - \theta \leq 0$, $g_i(x) \leq 0$. So WLOG $\min c^T x$ s.t. $g_i(x) \leq 0$, $x \in X$.

Lemma

$g_i \in C^1$ convex $\Rightarrow \{x : g_i(x) \leq 0 \ \forall i\} \subseteq S^k$ for all k (cuts never remove feasible points).

Algorithm

0. $k = 0$, $S^0 = \mathbb{R}^n$.
1. $x^k \in \arg \min c^T x$ s.t. $x \in S^k \cap X$.
2. If $g_i(x^k) \leq 0$ for all i : x^k is optimal, stop.
3. Pick $i \in \arg \max_i g_i(x^k)$; $S^{k+1} = S^k \cap \{x : g_i(x^k) + \nabla g_i(x^k)^T(x - x^k) \leq 0\}$; $k \leftarrow k + 1$, go to 1.

Theorem. X compact, $g_i \in C^1$ convex $\Rightarrow$ every limit point of $\{x^k\}$ is optimal.

Barrier and interior point methods

$\min f(x)$ s.t. $g_j(x) \leq 0$ ($j = 1, \dots, m$), $x \in X$.

Assumptions

(1) $f, g_j \in C^0$; (2) X closed; (3) $S = \{x \in X : g_j(x) < 0 \ \forall j\} \neq \emptyset$; (4) every feasible $x \notin S$ is a limit of points of S : for any $\delta > 0$ there is $\bar{x} \in S$ with $\|x - \bar{x}\| < \delta$.

Barrier subproblem

$$\min_{x \in S} f(x) + \varepsilon^k B(x), \quad \varepsilon^k > \varepsilon^{k+1} > 0, \ \varepsilon^k \rightarrow 0,$$

$$B(x) = -\sum_{j=1}^m \ln [-g_j(x)] \quad (\text{log barrier}), \quad B(x) = -\sum_{j=1}^m \frac{1}{g_j(x)} \quad (\text{inverse barrier}).$$

Log-barrier applied to LP

$$(P) \ z^* = \min c^T x \text{ s.t. } Ax = b, \ x \geq 0 \quad (D) \ \max b^T \lambda \text{ s.t. } A^T \lambda \leq c.$$

(equivalently (P): $\min -b^T \lambda$ s.t. $-A^T \lambda \geq -c$; (D): $\max -c^T x$ s.t. $Ax = b, \ x \geq 0$)

Central path

$$x(\varepsilon) \in \arg \min_{x \in S} F_\varepsilon(x), \quad F_\varepsilon(x) = c^T x - \varepsilon \sum_{i=1}^n \ln x_i, \quad S = \{x : Ax = b, \ x > 0\}.$$

Inner solver: constrained Newton = gradient projection with Hessian scaling

$$x \leftarrow x + \alpha(\tilde{x} - x), \quad \tilde{x} \in \arg \min_{Az=b} \nabla F_\varepsilon(x)^T(z - x) + \frac{1}{2}(z - x)^T \nabla^2 F_\varepsilon(x)(z - x).$$

Penalty and augmented Lagrangian methods (equality)

$$(P) \min f(x) \text{ s.t. } h(x) = 0, x \in X, \quad L_c(x, \lambda) = f(x) + \lambda^T h(x) + \frac{c}{2} \|h(x)\|^2.$$

Iterate: $x^k \in \arg \min_{x \in X} L_{c^k}(x, \lambda^k)$, then $\lambda^{k+1} = \lambda^k + c^k h(x^k)$.

Theorem

Let $X = \mathbb{R}^n$, $f, h \in C^1$, $\{\lambda^k\}$ bounded, $c^k \rightarrow \infty$. Suppose $\nabla_x L_{c^k}(x^k, \lambda^k) = 0$ for all k and $x^k \rightarrow x^*$ with $\nabla h(x^*)$ of rank m . Then

$$\lambda^k + c^k h(x^k) \rightarrow \lambda^*, \quad \nabla_x L(x^*, \lambda^*) = 0, \quad h(x^*) = 0.$$

Proof device (for convergence of penalty methods)

$$F^k(x) = f(x) + \frac{c^k}{2} \|h(x)\|^2 + \frac{\alpha}{2} \|x - x^*\|^2; \quad \nabla F^k = \nabla f + c^k \nabla h^T h + \alpha(x - x^*), \\ \nabla^2 F^k = \nabla^2 f + c^k \nabla h^T \nabla h + c^k \sum_i h_i \nabla^2 h_i + \alpha I.$$

Augmented Lagrangian with inequality constraints; remarks

Convert $g_j(x) \leq 0$ to $g_j(x) + z_j^2 = 0$:

$$L_c(x, z, \lambda, \mu) = f + \sum_i \lambda_i h_i + \frac{c}{2} \sum_i h_i^2 + \sum_j \mu_j (g_j + z_j^2) + \frac{c}{2} \sum_j (g_j + z_j^2)^2.$$

Minimizing over z in closed form ($\min_{x,z} L_c = \min_x \min_z L_c$):

$$L_c(x, \lambda, \mu) = f(x) + \lambda^T h(x) + \frac{c}{2} \|h(x)\|^2 + \sum_j \left[\mu_j g_j^+(x, \mu, c) + \frac{c}{2} (g_j^+(x, \mu, c))^2 \right], \quad g_j^+ = \max \left\{ g_j(x), -\frac{\mu_j}{c} \right\}.$$

Multiplier update: $\lambda^{k+1} = \lambda^k + c^k h(x^k)$ (and $\mu_j^{k+1} = \max\{0, \mu_j^k + c^k g_j(x^k)\}$).

Remarks

- ① As $c \rightarrow \infty$, $x^k \rightarrow x^*$ and $\lambda^k \rightarrow \lambda^*$ (optimal solutions).
- ② The condition number of $\nabla_{xx}^2 L_c$ grows with c .
- ③ Convergence of multipliers: $\frac{\|\lambda^{k+1} - \lambda^*\|}{\|\lambda^k - \lambda^*\|} \propto \frac{1}{c^k}$ (linear, superlinear if $c^k \rightarrow \infty$).